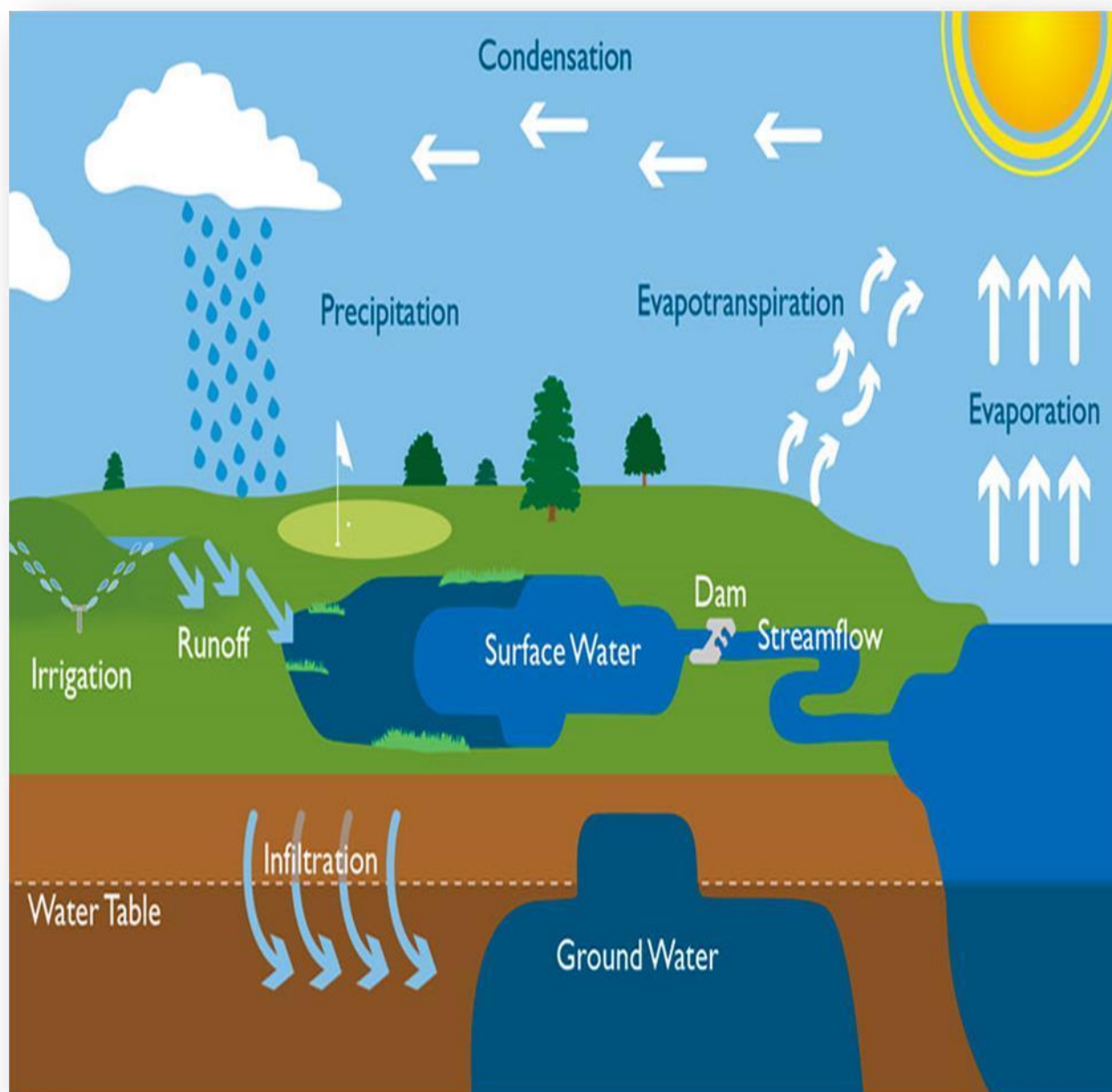


Ethiopia Meteorology Institute



Hydro Meteorological and flood monitoring Bulletin for Kiremt, 2026
impact assessment and Hydro meteorological impact outlook for Bega,
2026/27

Foreword

This seasonal hydro meteorological bulletin is prepared and disseminated by the Ethiopia Meteorological institution (EMI) of Ethiopia, for the purpose of providing hydro meteorological information to different sectors of the community involved in water related activities.

In general, Hydrometeorology is concerned with the study of the atmosphere and land phases of the hydrologic cycle, particularly, on the interrelationships involved. In this bulletin, more emphasis is given to presenting the results of analyses done on the extreme rainfall events as well as the moisture status prevailed over river catchments.

Accordingly, the data used in producing this bulletin are collected from selected indicative meteorological stations, which are believed to represent each of the main river catchments (hydrological regimes) of the country and the results of the hydro meteorological analyses are presented in maps format. Analysis presented in the forms of maps indicates comparisons of the total and extreme monthly rainfall events, monthly mean temperature and aridity index conditions for each basin.

Thus, the information contained in this bulletin is believed to be helpful in monitoring the performances of many hydraulic structures such as culverts, bridges, reservoir spillways, road embankments, dikes, flood prone areas as well as in planning and designing such new structures over the respective basins. It also gives the user an insight into the value as well as the contributions of the hydro-meteorological information towards the accomplishment of water resources assessment and management with respect to sustainable development of the country. Meanwhile, your comments and constructive suggestions are highly appreciated to make the objectives of this bulletin a success.

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I. Introduction

Ethiopia is located between latitudes of 3.8°N to 14.5°N and longitudes of 33°E to 48°E with an area of about 1.12 million km². The varied topography of the country shows extreme changes in altitude with its lowest point at about 120meters below sea level (Kobab Sink Afar depression) and its highest point about 4620 meters above sea level (Ras dashen). These physiographic variations create a large difference in meteorological and hydrological condition both by time and space.

From meteorological point of view, there are three seasons in Ethiopia; Belg, Kiremt and Bega.

Belg (February-May) is the small rainy season in Ethiopia. Much of the northeastern, central, southern, southwestern, eastern and southeastern parts of the country receive considerable amount of rainfall during this season.

Kiremt (June-September) is the main rainfall season for most parts of the country except for the lowlands of southern and southeastern Ethiopia.

Bega (October-January) is mostly a dry season for most parts of the country except for southwestern as well as the lowlands of south and southeast Ethiopia.

In general, the mean annual rainfall amount ranges from 2400mm (over south western) to 500 and below over the northeastern and southeastern lowlands. Hydro meteorologically a rainy day is considered as the one with 2.5 mm of rain or more but in this publication a rainy day is one regardless of the amount.

In Ethiopia, water resources availability in terms of space shows a marked discrepancy when one goes from east to west. The eastern part of the region compromise 7 catchments with only 11 percent of the water resource and while the west compromise 5 catchments with 89 percent of water resources.

II. Catchments profile

Catchment

Location

Mereb – Gash Catchment: -

Northwestern tip of Tigray.

Atbara-Tekeze Catchment: -

The Tekeze River basin is situated in the northwest of Ethiopia between 11° 04' N and 15° 02' N, and 36° 03' and 39° 50' E. It is bordered by the Mereb River basin and by Eritrea in the north, the Atbara River plains in Sudan in the west, the Abay River basin in the south and Danakil basin in the east.

Blue Nile/ Abbay Catchment: -

Roughly 130 N South of Gondar to 11° 30' N, and west of 39° 45' E of Wollo, northwestern parts of Shoa; Gojam except the South Western and Western narrow area, Wellega and extreme Eastern tip of Illubabor together with a narrow northeastern strip of Keffa. It is the largest catchment that covers about 16 percent of the total area of Ethiopia. The Catchment that includes the Lake Tana, Upper Abbay (to Guder confluence), Middle Abbay (to Didessa confluence), Didessa, Dabus, Lower Abbay, Dinder and Rahad Sub-basin.

Baro –Akobo Catchment: -

The south western and western narrow strip of Wellega, except the eastern tip, the whole of Illubabor and southwestern tip of Keffa. The Catchment has upper and lower sub-basins along Baro River. The Catchment It is the wettest catchment because of the highest rainfall over the area.

Danakil – Afar Catchment: -

East of 40° E of Tigray, North of 11° N of Wollo, narrow coastal strip south of 14° 30' N of Eritrea. The basin is the lowest region in the country where the kobar sink; with an elevation of about 120 meters b.s.l is found.

Awash Catchment: -

North of Garamuleta mountains, south of 11° 40' N of Wollo, south of 9° N of Shoa, Northern tip of Bale and North part of Arsi. The catchment has upper, middle and lower sub-catchments. In general the catchment is narrow at the upper part marked by

numerous volcanic mountains and wider at the lower part joining major tributaries from northwestern highlands and a number of seasonal wadies from the southeast highlands.

Gulf of Aden – Aysha Catchment): - Eastern narrow strip of Hararghe. It is a very dry area with no stream flow representative meteorological station. Thus, no assessment is done for this catchment in this publication.

Omo-Ghibe Catchment: -

Southwestern narrow strip of Shoa, the whole of Keffa except the southwestern tip, southwestern tip of Wellega, Western half of northern Omo and northwestern tip of Sidamo. The upper part of the catchment starts from the plateaus in north part of Ghibe and extends southward to the lower part of it (known as Omo River).

Central Lakes-Rift Valley Catchment: -

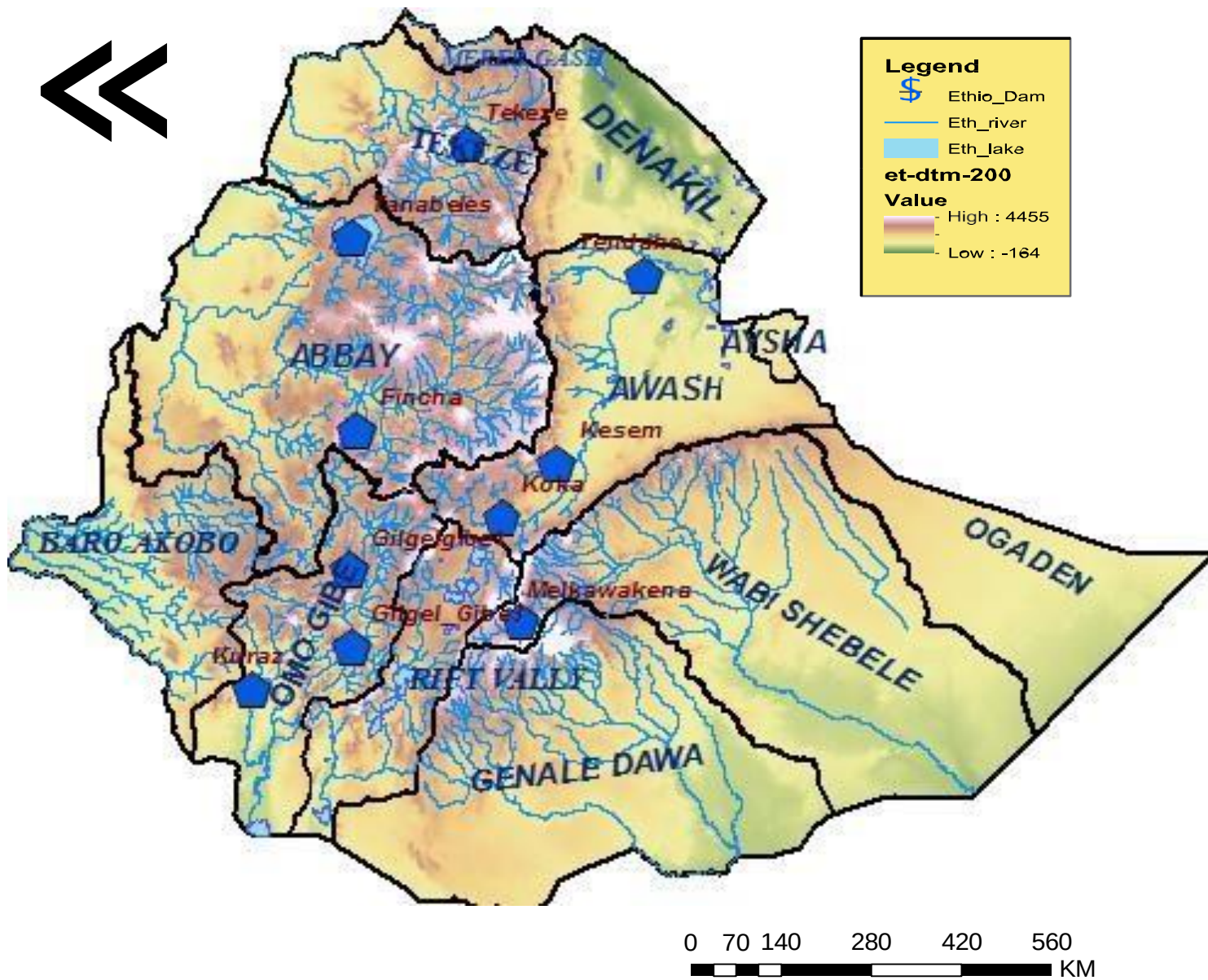
The whole of North and South Omo, west and southwestern narrow strip of Sidamo, southwestern portions of Shoa and western narrow tip of Bale and western part of Arsi. The catchment is found in the Great Rift Valley system and typically known by its lakes and streams. Lakes which adjoin the awash catchment are found in its upper part, while Lake Awassa and Bilate in its central part and end to chamo bahr in its lower part.

Genale Dawa Catchment: -

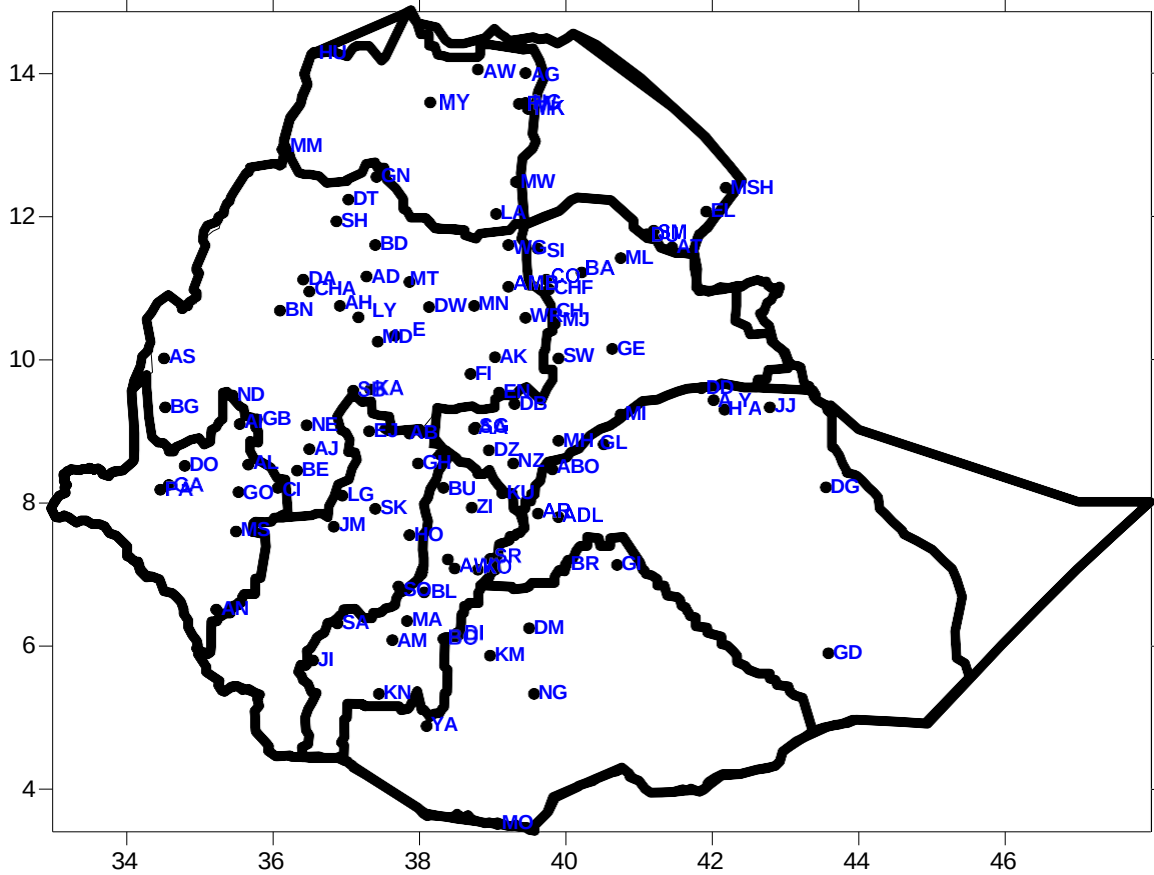
The western half of Bale (South of Goba) and southeast, southwestern and northeastern parts of Sidamo. The catchment constitute three river systems namely Dawa ,Genalle and Wabi Gestaro that meet each other before they cross the Ethio-Somalia border.

III. Major River Catchments in Ethiopia, Location and Spatial Status

No.	Catchement Name	Area (km ²)	Length in Kilo meter			Volume of water bm ³ /An num	Altitude (meter)
			Within Eth.	Outside Eth.	<i>Total</i>		<i>Peaks (Highest & Lowest)</i>
01	Mereb-Gash	5,700	440	160	600	0.15	North tip of Tigray
02	Tekaze – Atbar	90,001	608	560	1168	8.13	4620 Ras Dashen 125 Tikil -Dengay
03	Blue Nile(Abbay)	204,100	800	650	1450	52.62	4231 Guna 200 Horekelife
04	Baro - Akobo	75,912	227	280	557	23.55	3700 Masha 410 Jikawo
05	Afar (Denakil)	62,882	-	-	-	0.86	
06	Awash	112,696	1200	-	1200	4.6	4000 N.Shewa 4001 NW mt. 4002 of A.A 250 L.Abe
07	Aysha	2223				0.86	
08	Omo-Ghibe	78,213	760	-	760	17.96	4203Guge/Gurage Mt. 195 Chiri
09	Rift valley	54,900	-	-	-	5.63	
10	Genale - Dawa	171,042	480	570	1050	5.88	4310 Bale mt./Batu 500 Dolo Odo
11	Wabi - Shebele	205,697	1340	660	2000	3.16	3626 Mt.Gololcha 200 Somalia Desert
12	Ogaden	77,121	-	-	-	-	1500 Turkile 350 Gelad



V. Meteorological Station distribution used for hydro meteorological Bulletin.



STATION	CODE	STATION	CODE	STATION	CODE	STATION	CODE
A.A (Bole)	AA	Cheffa	CHF	Gonder	GN	Mille	ML
Abomsa	ABO	Chercher	CH	Gore	GO	Mira Abaya	MR
Adet	AD	Chira	CI	Hageremariam	HG	Motta	MT
Adigrat	AG	Combolcha	CO	Harer	HA	Moyalles	MO
Adwa	AW	Dangla	DA	Hossana	HO	Mytsebery	MY
Aira	AI	Debrebrhan	DB	Humera	HU	Nazaret	NZ
Alem ketema	AK	Degehabur	DG	Jijiga	JJ	Nedjo	ND
Alemaya	AY	Debre markose	DE	Jimma	JM	Negele	NG
Alge	AL	Debre Tadore	DT	Jinka	JI	Nekemt	NE
Ambamariam	AMB	Debre Zeit	DZ	Kachise	KA	Pawe	PA
Ambo	AB	Dembidolo	DO	Kibremengist	KM	Sawla	SA
Arbaminch	AM	Dilla	DI	Konso	KN	Sekoru	SK
Arjo	AJ	DireDawa	DD	Kulumsa	KU	Semera	SM
ArsiRobe	AR	Dolomena	DM	Koffele	KO	Freweyni	FW
Assaita	AT	Dubti	DU	Konso	KN	Shahura	SH
Assossa	AS	Ejaji	EJ	Kulumsa	KU	Shambu	SB
Awassa	AW	Elidar	EL	Lalibela	LA	ShewaRobit	SW
Ayehu	AH	Enewary	EN	Layber	LY	Shire	SR
Aman	AN	Elidar	EL	Limugenet	LG	SholaGebeya	SG
Bale Robe	BR	Enewary	EN	Maichew	MW	Sirinka	SI
BahidAr	BD	Fitche	FI	Mankush	MA	Sodo	SO
Bati	BA	Gambella	GA	Masha	MSH	WegelTena	WT

Beddele	BE	Gelemso	GL	Mehalmeda	MD	Wereillu	WR
Begi	BG	Gewane	GE	Mekaneselam	MN	Yabello	YB
Blate	BL	Ghion	GH	Mekele	MK	Ziway	ZY
Bui	BU	Gimbi	GB	Metehara	ME		
Bullen	BN	Ginir	GI	Meisso	MS		
Chagni	CG	Gode	GD	Metema	MM		

The above stations have five basic meteorological elements they send daily records for Addis Ababa main office of EMI. We use the meteorological elements which are the main factors for hydro meteorological impacts. These are rainfall, temperature, wind speed, evaporation and sunshine duration. This information is important to guide for different water resource activities.

1. Kiremt season description

Kiremt from June to September is mostly wet to very wet season for most parts of the river basin except south and south eastern basins such as lower and middle Wabi shebele, lower Rift valley, lower Omo Gibe, middle and lower Genale Dawa catchments. It has significant important for water resource of the country. The main source of runoff is Kiremt seasonal rainfall, which occur high flow period and maximum availability of surface and ground water over those catchments. It is known that Kiremt season rainfall covers the water demand of all reservoir and river basin water storage. Dam, reservoir and ponds can capture the highest volume of water during this season. The occurrence of flood and land slide is widely known phenomenon over flood prone area and rugged surface of the country during Kiremt season respectively.

Methods

To compute the aridity index, we use Thornthwait method, which is computed from the monthly values of rainfall and evaporation. The evaporation is computed empirically from mean monthly air temperature. In assessing the effectiveness of rainfall, in terms of water availability relationships between the rainfall and air temperature has been worked out in terms of moisture indices. The aridity index values above 350 which shaded in Blue green very wet and 128 were show wet condition. Light green humid, yellow sub- humid and red values show dry condition.

Where RF= monthly rainfall in mm; T = monthly mean temperature in °C

2. Aridity status for Kiremt 2026 at different river basins

During the 2026 Kiremt season, rainfall patterns evolved significantly across Ethiopia's primary river basins, driving substantial surface water flows and distinct spatial shifts in moisture distribution. In June, the onset of the season established moderate to high surface water flows across western and central basins, notably within the Abay, Baro-Akobo, middle and lower Tekeze, upper and middle Omo-Gibe, Central Rift Valley, upper Awash, and upper Genale Dawa catchments. By July, hydrological conditions expanded northward and eastward into main Kiremt-dependent regions, resulting in wet to hyper-wet conditions across the Abay, Tekeze, upper Afar-Denakil, upper and middle Awash, Omo-Gibe, Central Rift Valley, upper Wabe Shebelle, and Genale Dawa basins. August marked the seasonal peak, generating widespread wet to very wet conditions across most major catchments, with hyper-wet conditions recorded in the Abay, Tekeze, Baro-Akobo, Omo-Gibe, and upper/middle Awash basins. Finally, in September, the primary moisture axis gradually retreated from the east and northeast toward the southwest and west, maintaining humid to very wet conditions across the Abay, Baro-Akobo, Omo-Gibe, upper Wabe Shebelle, and upper Genale Dawa basins, alongside localized areas in Afar-Denakil, as the main rainy season concluded. Shown below in figure 1

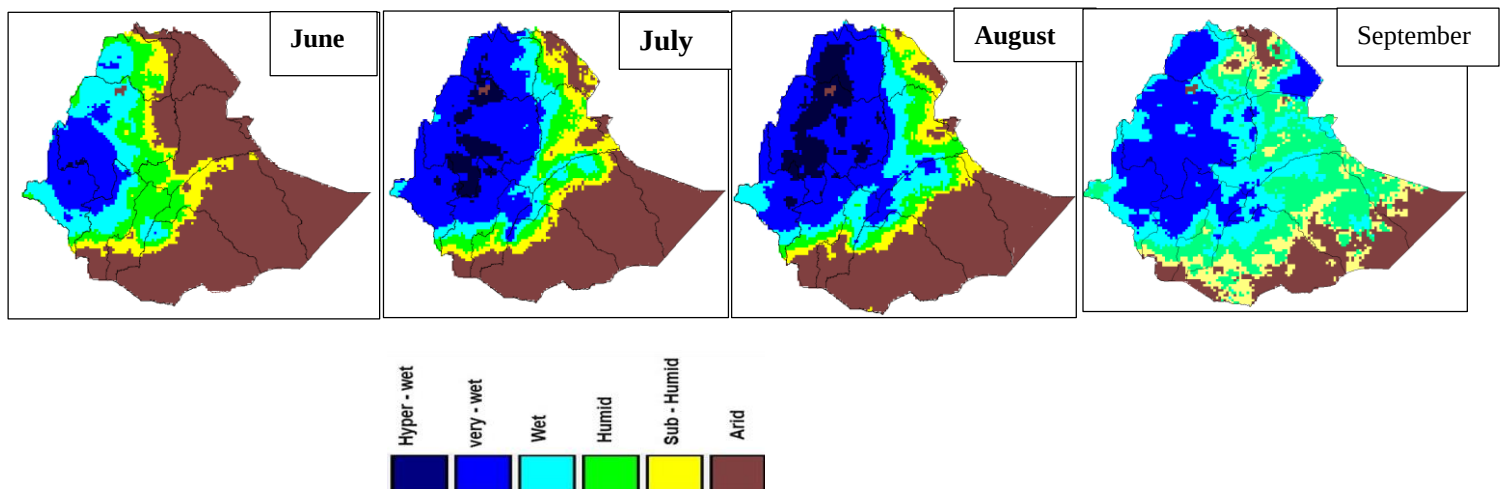


Figure 1 Aridity Index during Kiremt 2026

Distribution of heavy fall days exceeding 30mm during June to September 2026 over different river basins.

During the 2026 Kiremt season, heavy precipitation events were recorded across several major river basins, demonstrating varying monthly spatial distributions and peak frequencies. In June, heavy rainfall days were predominantly observed over the Abay, Baro-Akobo, Omo-Gibe, and Rift Valley catchments, with the Abay basin recording the highest frequency at six days. Heavy falls expanded significantly throughout July, most monitoring stations across the Abay, Baro-Akobo, Tekeze, Omo-Gibe, upper and middle Awash, and select Afar-Denakil catchments, where the Abay basin experienced a seasonal peak of four heavy rainfall days. By August, heavy precipitation remained concentrated across the majority of the Abay, Baro-Akobo, Tekeze, Omo-Gibe, and upper Central Rift Valley basins. Finally, in September, heavy fall occurrences persisted across most Abay, Baro-Akobo, and Omo-Gibe stations, as well as the upper and middle Tekeze basins, with the highest frequency reaching four days.

Kiremt (June to September) 2026

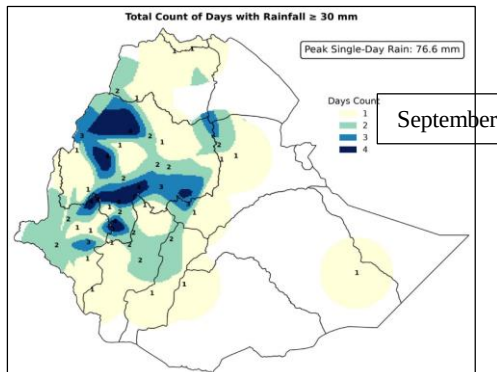
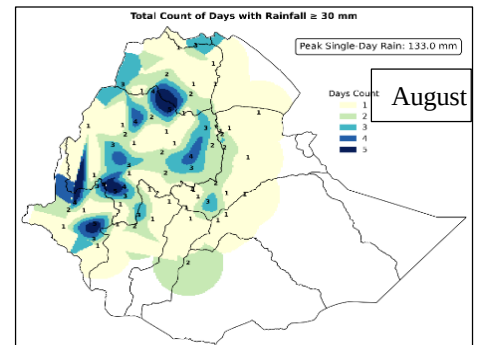
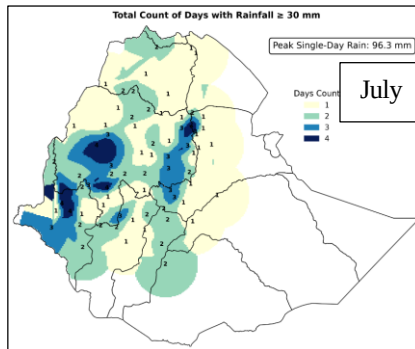
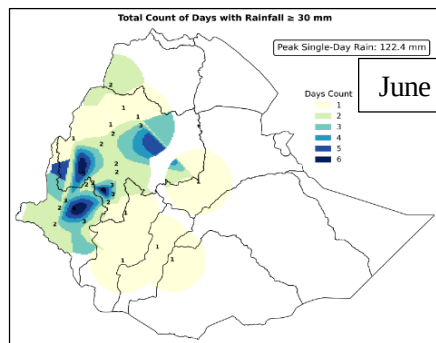


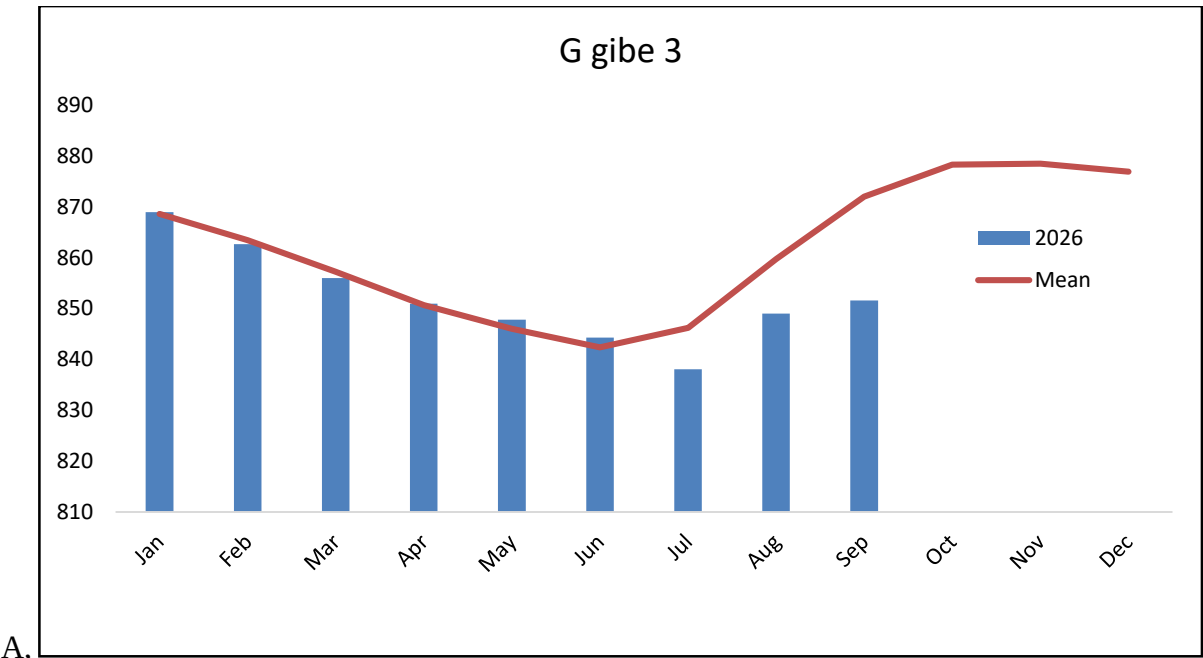
Figure 2 Distribution of Heavy fall from June to September, 2026.

Performance of Dams and Reservoirs water level in Kiremt 2026 season compared to long mean

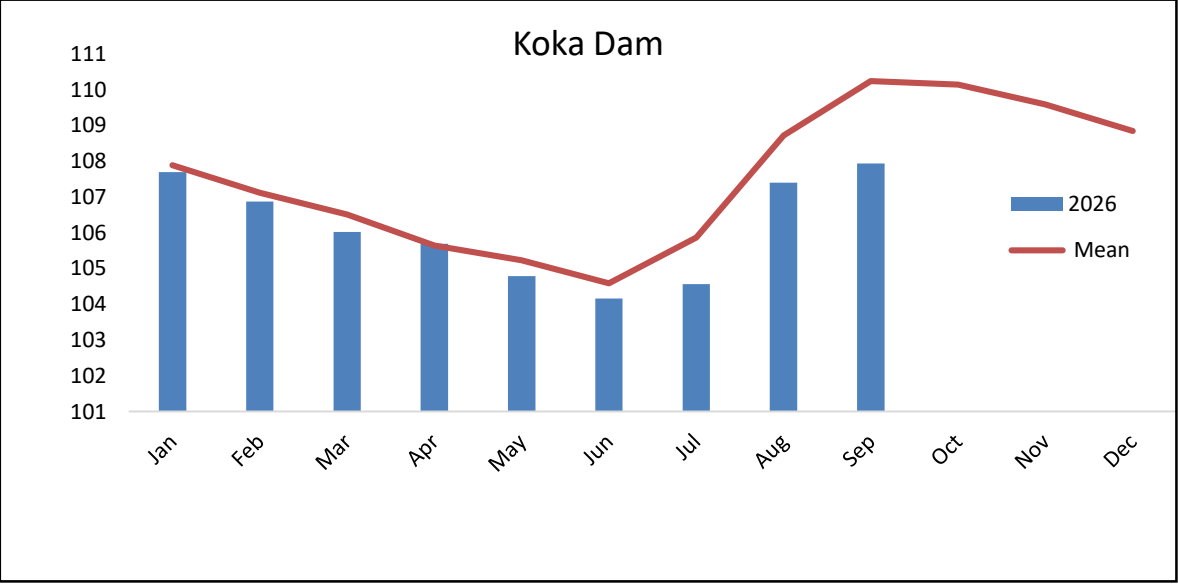
Rain-induced surface runoff during the Kiremt season serves as the principal driver of country of Ethiopia water storage, sustaining natural river systems, reservoirs, and artificial water bodies. Assessment of the completed 2026 Kiremt season confirms generally favorable precipitation patterns, which yielded strong hydrological performance and robust water accumulation across the majority of national river basins. However, localized storage deficits persisted, most notably in the Gil gel Gibe III and Koka reservoir catchments, where cumulative seasonal inflows remained significantly below their long-term mean averages. To mitigate the impact of these identified storage shortfalls, targeted upstream catchment restoration and soil-water conservation efforts are required across the Omo-Gibe and Awash River basins to improve base flow retention, reduce siltation, and build hydrologic resilience for future rainy seasons.

(4a-h).

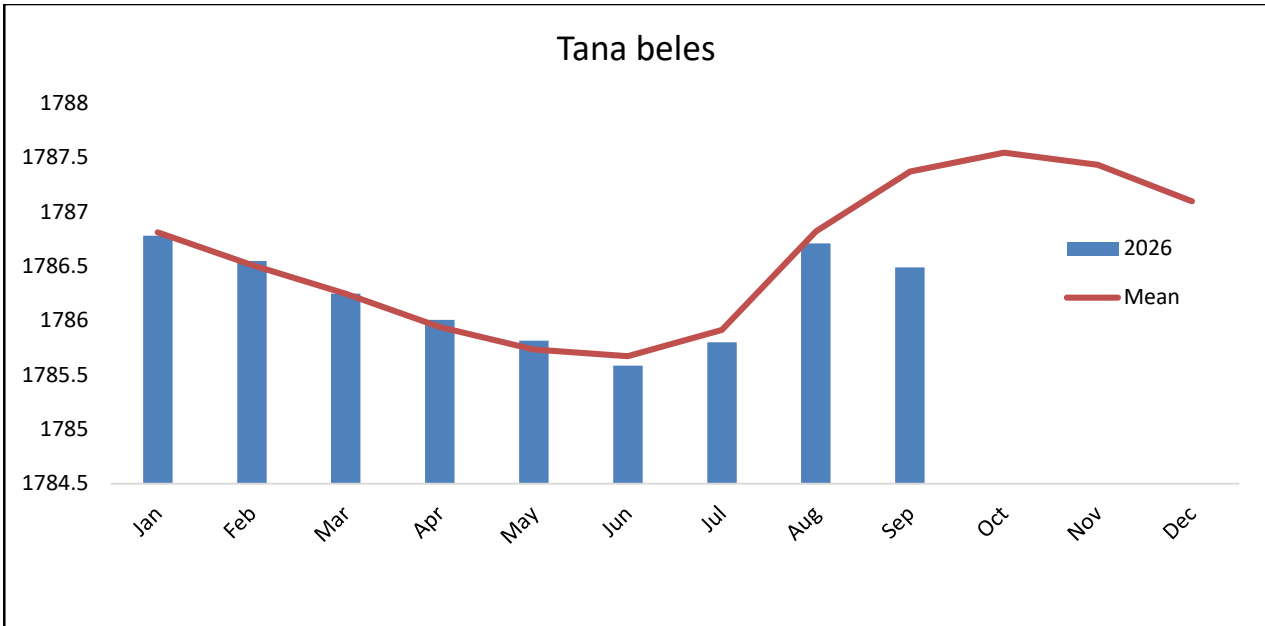
Data source: EPA (NGCC)



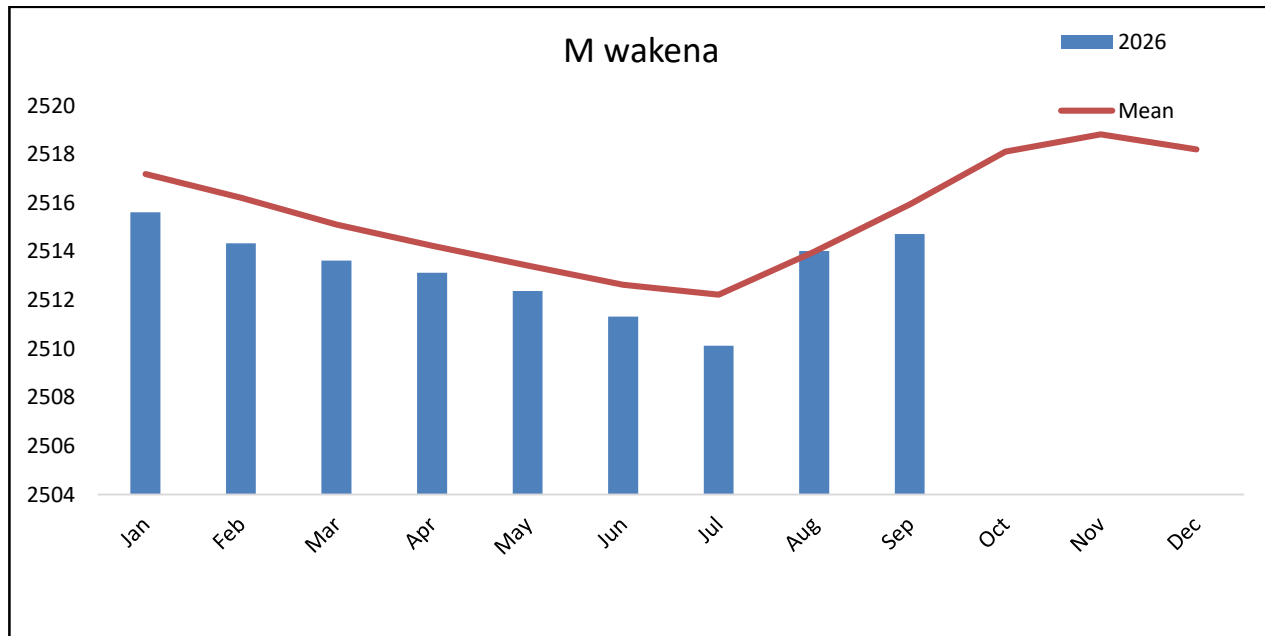
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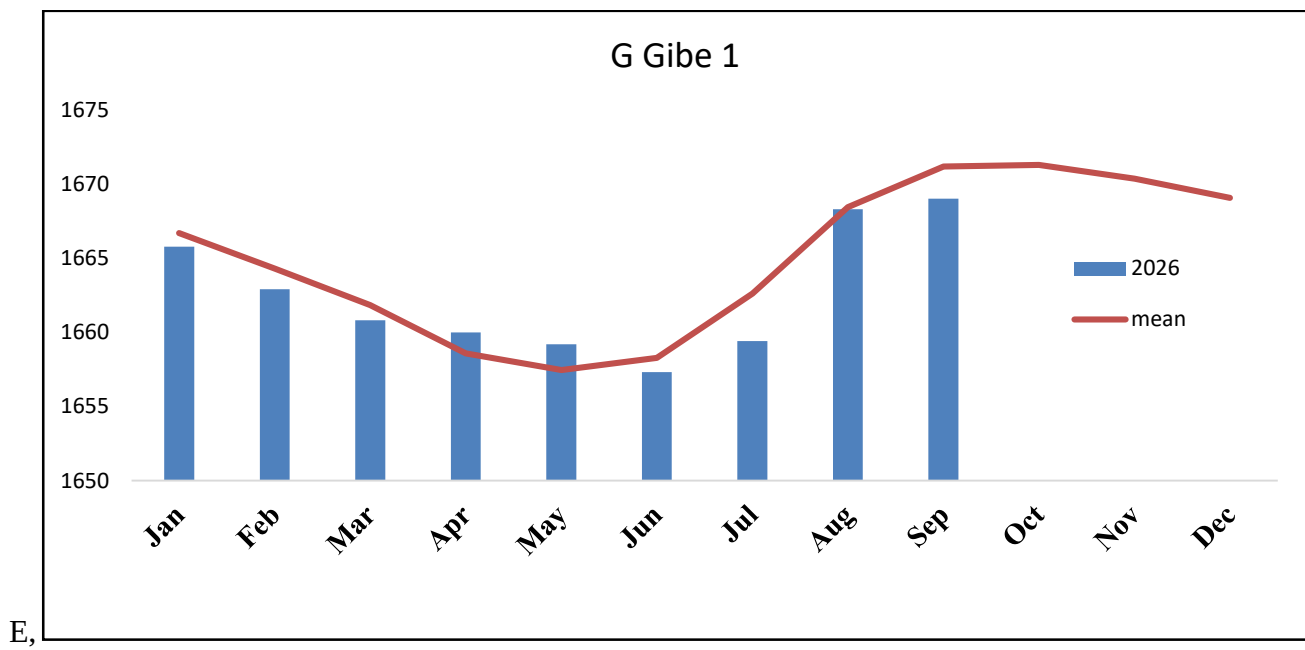


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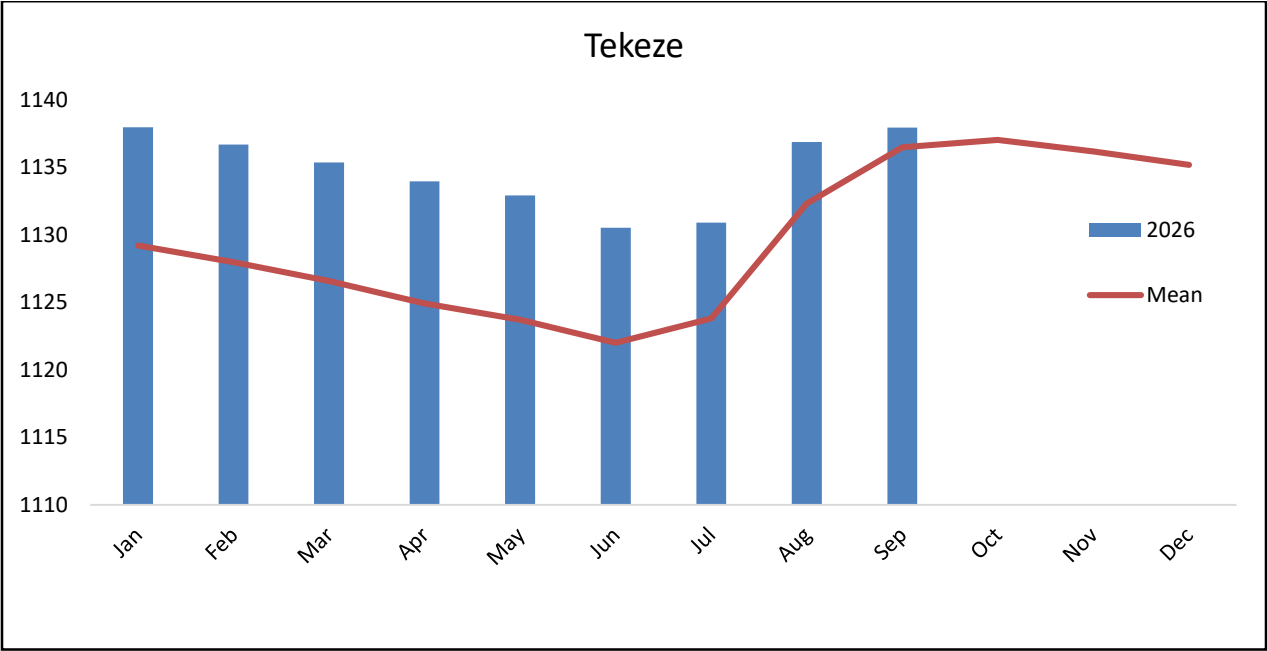


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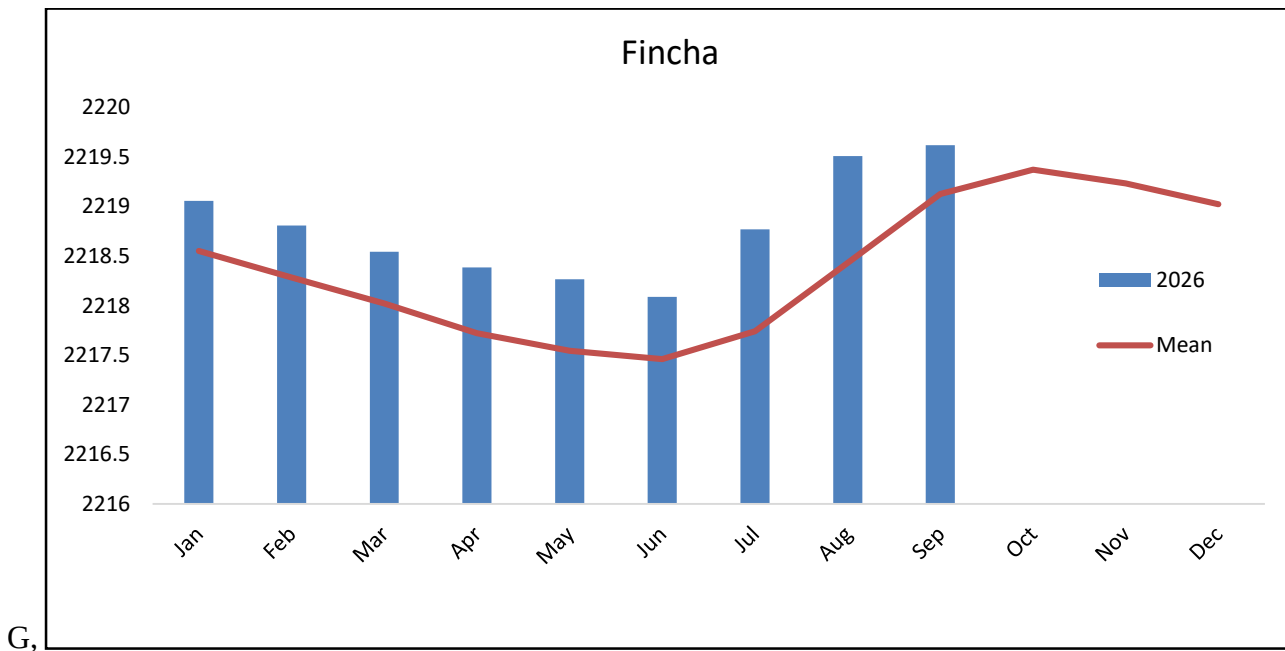


Figure 4a-h Water level of different dams.

Summary

During the 2026 Kiremt (June-September) rainy season, Ethiopia experienced favorable moisture conditions and strong surface water accumulation across most river basins. The season progressed from initial runoff in June to widespread wet and hyper-wet conditions during July and August particularly across the Abay, Tekeze, Baro-Akobo, Omo-Gibe, and upper and middle Awash catchments before retreating toward the southwest in September. Heavy precipitation events (≥ 30 mm/day) peaked at up to 6 days per month, with a seasonal single-day rainfall maximum of 133.0 mm and 130 mm recorded at Masha station in the Baro-Akobo basin.

Overall, rain-induced runoff significantly boosted natural river flows, reservoir levels, and groundwater recharge compared to long-term averages. However, localized storage shortfalls persisted in the Gilgel Gibe III and Koka reservoirs due to below-average cumulative seasonal inflows. To address these deficits and reduce siltation, targeted upstream soil water conservation and catchment restoration efforts are required in the Omo-Gibe and Awash River basins.

Conclusion

In conclusion, the 2026 Kiremt rainy season recorded favorable hydro-meteorological performance, bringing above-average moisture conditions and enhanced water availability across Ethiopia's major river catchments. The overall water levels in key dams and reservoirs performed better than long-term historical means, supported by widespread wet to hyper-wet conditions during the peak months of July and August across southern, southwestern, central, western, and northeastern regions. Although localized storage deficits persisted in specific catchments such as Gilgel Gibe III and Koka due to below-average cumulative inflows, the season successfully fulfilled water storage demands and boosted surface and groundwater resources across the vast majority of the country.

Hydro-meteorological impact outlook for Bega 2026-27 over different river basins

1. Introduction

Bega from October to January is mostly an arid season for most part of the river basin except Ogaden, Lower and middle Wabe shebele, lower Omo Gibe middle and lower Genale Dawa as well as lower central rift valley catchments. The main rainy month Bega season is October and November provides rainfall mainly for southern, south eastern and south western catchments. Sunny and windy condition dominated across many river basin thus it increase the loss of water by Evaporation. Availability of water is also decrease across upstream of main river basin. In some years' unseasonal rain slightly fevered for water availability.

2. Selected analogue Year

For the coming Bega season the selected analogue year **1997/98 and 2023/24** Were compared based on probabilistic seasonal forecast for **Bega 2026/27** and viewed out on catchments based map using geostatistical kriging method. Among those **1997/98** is the best analogue years that can inform the Aridity status in the coming Bega season.

3. Methodology

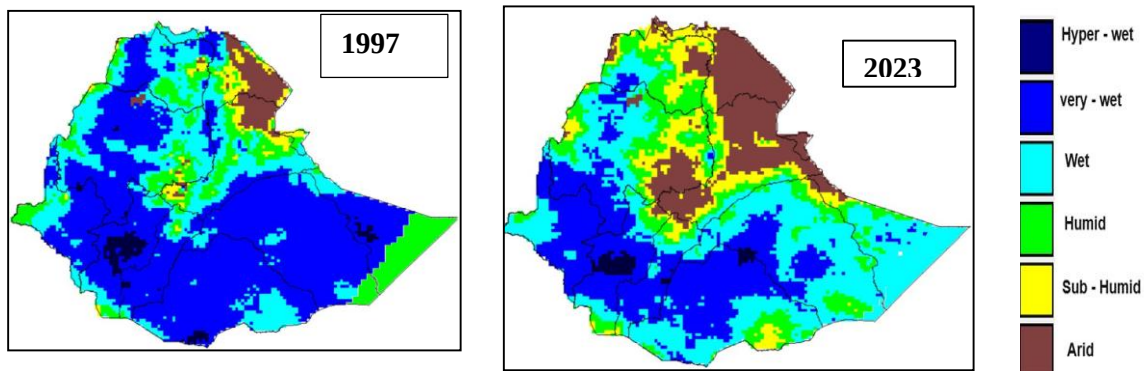
Thornthwaite introduced the concept of the precipitation effectiveness index, which is computed from the monthly values of rainfall and evaporation. The evaporation is computed empirically from mean monthly air temperature. In assessing the effectiveness of rainfall, in terms of water availability relationships between the rainfall and air temperature has been worked out in terms of moisture indices.

Where, R_f = monthly rainfall in mm;

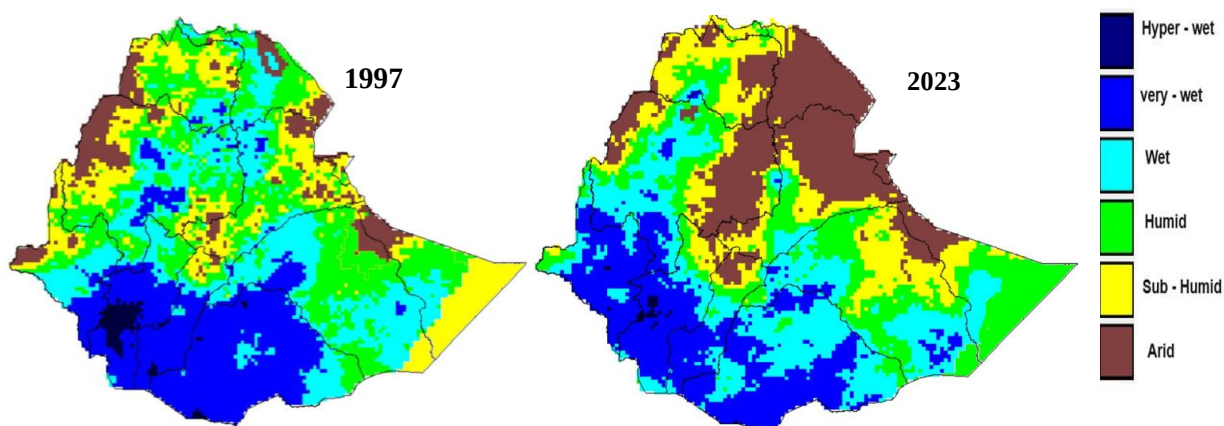
T = mean monthly temperature in $^{\circ}\text{C}$

October: Based on analysis of the two analogue years (1997 and 2023), catchments benefiting from Bega moisture are expected to begin accumulating initial surface water across most river basins in October.

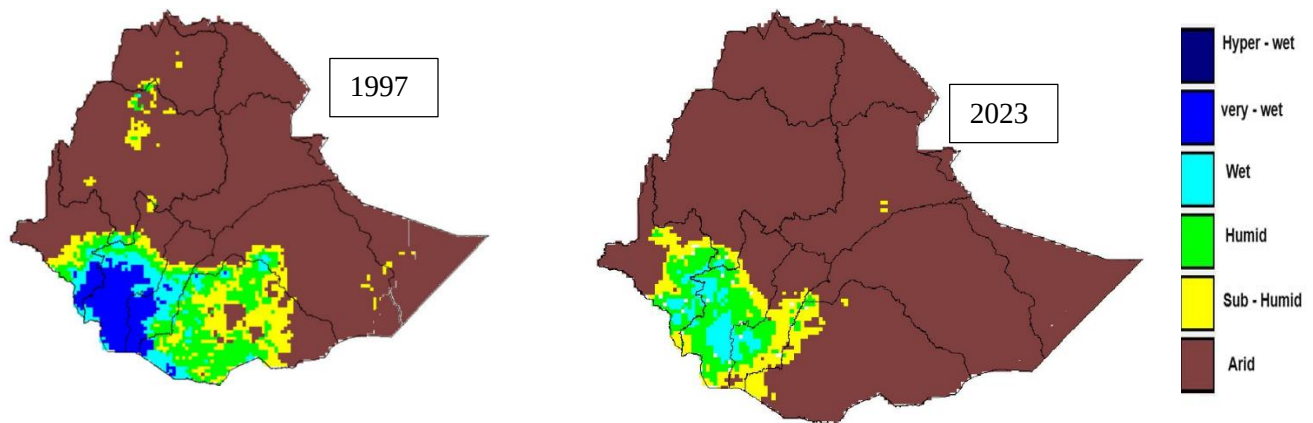
Most notably, Bega-benefiting basins specifically Baro-Akobo, Wabe Shebelle, Genale Dawa, and Ogaden alongside the Abay basin, are projected to experience predominantly humid to wet conditions throughout the month.



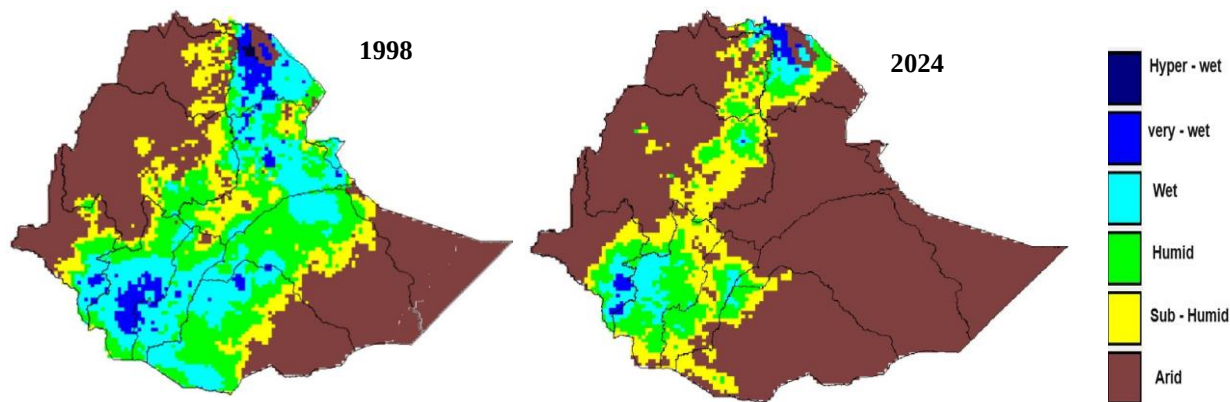
November: In this month, Based on analysis of the two analogue years (1997 and 2023), Bega moisture-benefiting catchments are expected to continue receiving significant surface water across most southern, southwestern, and southeastern basins in November. Specifically, Bega-benefiting basins such as Genale Dawa, Wabe Shebelle, Baro-Akobo, and Ogaden along with parts of the Abay and Rift Valley basins, are projected to experience predominantly wet to very wet hydrological conditions, while northern and northeastern areas transition toward drier conditions.



December: - Moisture distribution is expected to continue across the major catchments, benefiting Bega. The hydro meteorological conditions are expected to be sub-humid to wet over most parts of Omo-Gibe, Rift Valley, Baro-Akobo, and Genale Dawa, Wabe Shebelle and Abay basins and the middle and lower Ogaden catchments. However, the remainder of the country is expected to be dominated by arid to sub-humid conditions particularly affecting the Afar-Denakil, Aysha, middle and upper Abay, Tekeze and middle and lower Awash basins.



January:- Based on the analog years selected (1998 and 2024), Outlooks indicate that sub-humid to wet moisture conditions are likely to continue to dominate the major southern and southwestern catchments during the month of January of the upcoming 2026/27 Bega season, which includes most of the Omo-Gibe, Rift Valley, and upper Genale Dawa basins. In contrast, the rest of the country is expected to be dry and arid which will have great impact on the hydrological performance of most of the Awash, Wabe Shebelle, Abay, Ogaden, Aysha, Mereb-Gash and lower Baro-Akobo basins.



Hydro-Meteorological Summary and advisory for coming Bega (ONDJ), 2026-27

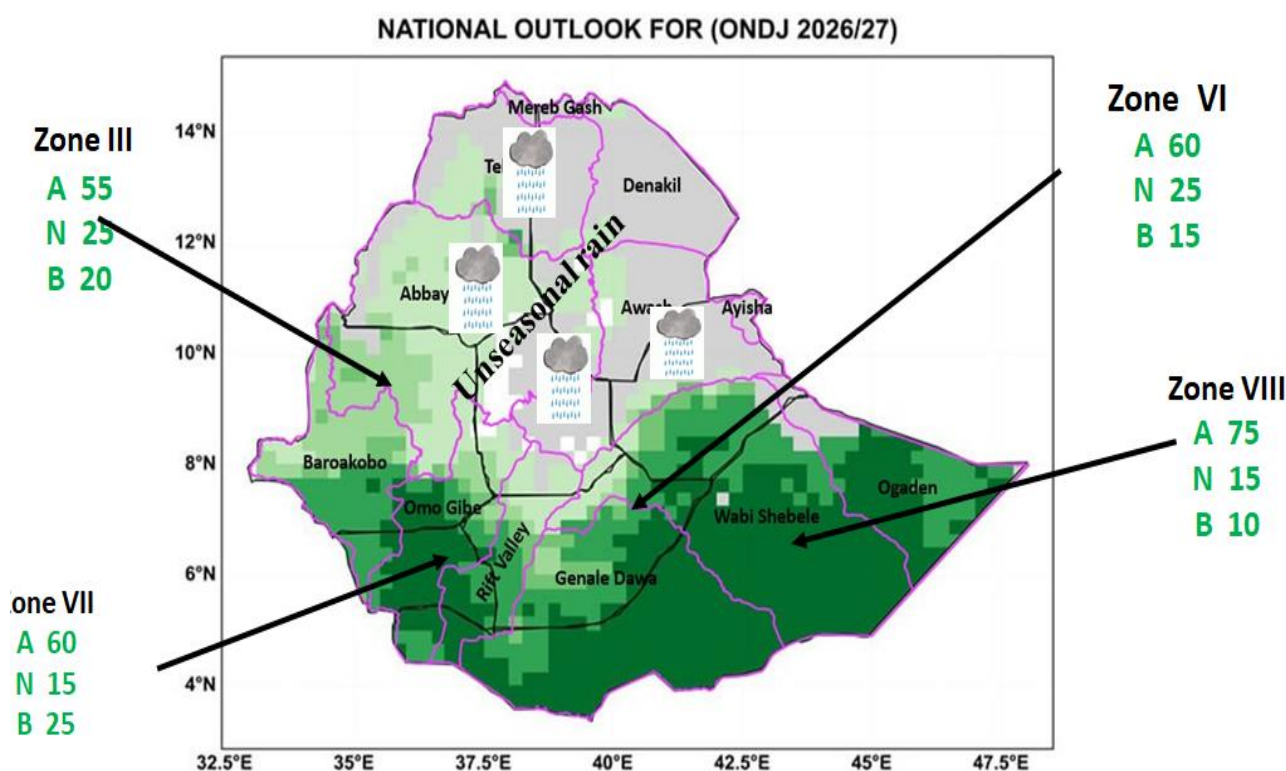
Summary

October: Humid to wet conditions across southern and southwestern catchments Baro-Akobo, Wabi Shebelle, Genale Dawa, Ogaden, and Abay support initial surface water accumulation.

November: Moisture shifts southward; wet to very wet conditions continue in Genale Dawa, Wabi Shebelle, Baro-Akobo, Ogaden, and parts of Abay Rift Valley, while northern areas dry out.

December: Dry to arid conditions dominate northern, central, and eastern basins Afar-Denakil, Aysha, middle upper Abay, Tekeze, and Awash. Sub-humid to wet levels persist in Omo-Gibe, Rift Valley, Baro-Akobo, Genale Dawa, Wabi Shebelle, and middle lower Ogaden.

January: Arid conditions expand across most northern, central, and eastern catchments. Sub-humid to wet conditions remain confined to Omo-Gibe, Rift Valley, and upper Genale Dawa.



Positive Impacts

- ❖ **Enhanced Rainfall in Bega-Benefiting Basins:** High probabilities of Above-Normal rainfall across Ogaden, Wabi Shebelle, Genale Dawa, Omo Gibe, Rift Valley Baro-Akobo, and western Abbay will significantly bolster surface water flow and soil moisture levels.
- ❖ **Water Storage & Capacity Optimization:** Favorable seasonal rainfall over southern and southeastern catchments will boost reservoir inflows, pond retention, and recharge groundwater aquifers during the dry period.
- ❖ **Potential Inflow Boost from Unseasonal Rains:** Areas in the central and northern basins including Abbay, Tekeze, Awash, Denakil, and Mereb Gash are projected to receive unseasonal rainfall, providing unexpected moisture support and reducing seasonal water stress.
- ❖ **Sustained Ecosystem & Agricultural Water Availability:** Strong rainfall probabilities across pastoral and agro-pastoral areas in the south and southeast will improve pasture regeneration and livestock drinking water supplies.

Negative Impacts

- ❖ **Risk of Localized Floorings & Flash Floods:** Extremely high probabilities of above-normal precipitation in Ogaden, Wabi Shebelle and Genale Dawa, Omo Gibe raise the risk of flash flooding, river overflow, and soil erosion in lowland flood-prone areas.
- ❖ **Impact of Unseasonal Rain on Harvests:** Unseasonal rainfall across northern and central catchments Abbay, Tekeze, and Awash during the crop harvesting phase could disrupt post-harvest agricultural activities and cause crop damage if not properly managed.
- ❖ **Elevated Evapotranspiration:** Intermittent warm days during the Bega season may still accelerate evaporative losses from open reservoirs and shallow ponds despite improved rainfall coverage.

Advisories and Recommendations

- ❖ **Optimize Rainwater Harvesting & Reservoir Storage:** Capitalize on the high likelihood of above-normal rainfall across southern and southeastern catchments by maximizing water harvesting in ponds, check dams, and small Dams.
- ❖ **Flood Mitigation & Infrastructure Monitoring:** Disaster risk management bodies and water authorities should prepare flood protection infrastructure, clear drainage channels, and monitor river levels along vulnerable segments of the Wabi Shebelle, Genale Dawa, and Omo Gibe basins.
- ❖ **Harvest Protection against Unseasonal Rain:** Farmers and agricultural operators in central and

northern regions expecting unseasonal rains should expedite crop harvesting and secure crop storage facilities to prevent moisture damage.

- ❖ Practical Hydropower & Irrigation Management: Coordinate dam dispatch strategies to balance water storage benefits with flood control margins across major reservoirs.
- ❖ Active Information Monitoring: Continuously track updated short-range and daily advisories issued by the Ethiopian Meteorology Institute (EMI) via official portals: <http://www.ethiomet.gov.et> or <http://www.Ethiopia.ranet.net>.